

Terminal 1

```
source /opt/ros/jazzy/setup.bash

source ~/workspace/ros2_kortex_ws/install/setup.bash

export LIBGL_ALWAYS_SOFTWARE=1

export GZ_SIM_RESOURCE_PATH=$(dirname $(ros2 pkg prefix --share
kortex_description)):GZ_SIM_RESOURCE_PATH

export LD_LIBRARY_PATH=/opt/ros/jazzy/lib:$LD_LIBRARY_PATH

export
GZ_SIM_SYSTEM_PLUGIN_PATH=/opt/ros/jazzy/lib:$GZ_SIM_SYSTEM_PLUGIN_PATH

gz sim -r gen3_world.sdf
```

Terminal 2

```
source /opt/ros/jazzy/setup.bash

ros2 run ros_gz_bridge parameter_bridge /clock@rosgraph\_msgs/msg/Clock\[gz.msgs.Clock
```

Terminal 3

```
ros2 launch my_robot_task_api rsp.launch.py
```

Terminal 4

```
ros2 run ros_gz_sim create -name gen3_lite -topic /robot_description
```

Terminal 5

```
ros2 run controller_manager spawner joint_state_broadcaster --controller-manager
/controller_manager

ros2 run controller_manager spawner joint_trajectory_controller --controller-manager
/controller_manager

ros2 run controller_manager spawner gen3_lite_2f_gripper_controller --controller-manager
/controller_manager
```

Terminal 6

```
ros2 run ros_gz_bridge parameter_bridge \

/rgbd_camera/image@sensor_msgs/msg/Image[gz.msgs.Image \

/rgbd_camera/depth_image@sensor_msgs/msg/Image[gz.msgs.Image \

/rgbd_camera/camera_info@sensor_msgs/msg/CameraInfo[gz.msgs.CameraInfo \
```

Terminal 7

```
python ~/workspace/my_robot_task_api/my_robot_task_api/yolo_detector_node.py --ros-args -p model_path:=/home/rhenius/workspace/my_robot_task_api/models/sim_best.pt -p conf:=0.35 -p image_topic:=/rgbd_camera/image
```

--requires python environment

Terminal 8

```
python3 ~/workspace/my_robot_task_api/my_robot_task_api/bbox_to_pose_node.py \
--ros-args -p camera_frame:=rgbd_camera_optical_frame -p target_frame:=base_link
```

Terminal 9

```
python3 ~/workspace/my_robot_task_api/my_robot_task_api/llm_planner_improved.py
```

--requires python environment

Terminal 10

```
python3
~/workspace/my_robot_task_api/my_robot_task_api/improved_move_group_executor_node.py
```

Terminal 11

```
ros2 topic pub /task_command std_msgs/String "{data: 'pick the box and place it on the tray'}" -l
```